BT5153: Applied Machine Learning for Business Analytics

Assignment 3

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Task 1: Guided Evaluation

1.1 Built the Test Set

Q1: What datasets were used to train the AI text detection models?
Ground Truth Answer: They used dataset from Kaggle (contains 29,145 samples of student essays and GPT(Curie)-generated essays focused on the topic of car-free cities), HuggingFace (consists of 150,000 pairs of Wikipedia introductions and their AI-generated versions) and 6 prestigious essay competitions (includes 50 past winning essays from five global competitions, matched with GPT-3.5/4.0 generated content of similar lengths, averaging about 1000 words per essay).
Expected Passage: Textual data from Kaggle, Hugging Face and six prestigious essay competitions were collected and processed to support the development and evaluation of machine learning models.

Q2: What are the four features extracted for AI-generated text detection?
Ground Truth Answer: According to the document, the four key features extracted from the pre-processed text data for AI-generated text detection are Sentence length variation (standard deviation), Mean word length, Percentage of stop words per text (word count), Percentage of adjectives per text (word count).
Expected Passage: In the feature engineering phase, drawing insights from the exploratory data analysis (EDA), four key features are extracted from the pre-processed text data to enrich the dataset and provide insightful analysis for AI-generated text detection: 1. Sentence length variation (standard deviation): This feature signifies the variation in sentence complexity and language proficiency within each text sample. 2. Mean word length: Providing an indication of the complexity of word usage. Detection of AI-Generated Text for Essay Competitions 3. Percentage of stop words per text (word count): Reflecting sentence structure, syntax, and language fluency, this feature offers valuable insights into the composition of the text. 4. Percentage of adjectives per text (word count): This feature serves as an indicator of rich description, expressive language, and subjective interpretation within the text samples.

Q3: What is DistilBERT and why was it chosen for this task?
Ground Truth Answer: DistilBERT is a compact and efficient variant of BERT created through knowledge distillation, retaining approximately 97% of the original model's performance despite having 40% fewer parameters . It was selected for this project because its reduced size allows for faster inference, efficient memory usage, and easy deployment on systems with limited computational resources without significantly compromising performance.
Expected Passage: DistilBERT, a compact and efficient variant of BERT, is chosen as a suitable model for detecting AI-generated essays and preventing plagiarism in

competitions. BERT, renowned for its advanced language understanding capabilities, employs bidirectional context and a transformer architecture to comprehend word meaning within sentences. DistilBERT is developed through a process known as knowledge distillation, where a smaller model is trained to replicate the behavior of the larger BERT model. Despite having 40% fewer parameters, DistilBERT retains approximately 97% of BERT's performance on various benchmark tasks. This reduced model size enables faster inference and efficient memory usage, making DistilBERT an optimal choice for detecting AI-generated text and identifying instances of plagiarism in essay competitions. Furthermore, DistilBERT offers scalability advantages, as it can be easily deployed and utilized on systems with limited computational resources or memory availability, without compromising significantly on performance.

Q4: What is XLNet and how does it differ from traditional autoregressive models?
Ground Truth Answer: XLNet is a model specifically designed for language understanding tasks that serves as a strong candidate for detecting AI-generated content . It differs from traditional autoregressive models by employing a permutation language modeling objective that enables bidirectional context learning and considers dependencies among all input token permutations, thereby overcoming the limitations of unidirectional pretraining.
Expected Passage: XLNet, specifically designed for language understanding tasks is also a suitable candidate for detecting AI generated content. Unlike traditional autoregressive models, XLNet employs a permutation language modelling objective that allows for consideration of dependencies among all permutations of input tokens and bidirectional context learning, overcoming limitations related to unidirectional pretraining. This innovative approach enables XLNet to capture long-range dependencies between words in both forward and backward directions, leading to enhanced contextual understanding and improved performance on a wide range of natural language processing tasks. Moreover, XLNet's robustness to out-of-distribution text, achieved through permutation language modeling, bidirectional context understanding, and adaptive learning, allows it to mitigate biases present in AI-generated content. The flexibility and effectiveness of XLNet make it a promising choice for developing robust and accurate AI-generated text detection systems tailored for academic integrity applications

Q5: What preprocessing steps were applied to the text data?
Ground Truth Answer: The pre-processing stage began by removing duplicate entries and excluding any text incorrectly labeled simultaneously as both AI-generated and non-AI-generated . Following feature extraction, the text was converted to lowercase, stripped of punctuation and common stop words to reduce noise, and cleared of any extra spaces to ensure uniformity.
Expected Passage: In the data pre-processing stage, several essential steps are implemented to clean and prepare the dataset for AI-generated text detection. Firstly, duplicate entries are removed to ensure data integrity and eliminate redundancy.

Text entries incorrectly labeled as both AI-generated (1) and non-AI-generated (0) are excluded to maintain consistent labeling.

Q6: What were the precision and recall of the best DistilBERT configuration after domain adaptation?
Ground Truth Answer: After domain adaptation, the best DistilBERT configuration achieved a perfect precision score of 1, indicating flawless identification of AI-generated essays . Furthermore, the model successfully maintained a high recall score of 0.9, which demonstrated the overall effectiveness of this specific fine-tuning phase.
Expected Passage: The comparison of model 4's performance before and after domain adaptation is summarized in Table 2. Following additional fine-tuning on the 60 rows of 1000-word essays, precision further increased to 1 indicating perfect precision in detecting AI-generated essay. Moreover, the model maintained a high recall of 0.9. This improvement highlights the effectiveness of domain adaptation in enhancing the model's performance within the specific domain of academic essays.

Q7: What is the purpose of the two-phase ne-tuning approach?
Ground Truth Answer: The initial preliminary phase of fine-tuning optimizes model adaptability by determining the best dataset and meta-feature combinations, which helps overcome data size limitations by diversifying the models' exposure to various linguistic patterns and topics . The subsequent domain adaptation phase further refines the top-performing models using specific academic essay data, enabling them to capture the unique stylistic and structural characteristics common in essay competitions.
Expected Passage: Once the baseline performance metrics are established, each of the three pretrained models undergoes hyperparameter fine-tuning, aiming to minimize validation loss following a 7-3 train-test split. Each model is finetuned across four scenarios, comprising combinations of the Kaggle dataset and the hybrid dataset train_v2, with and without the inclusion of four meta-features. The optimal dataset and feature combination for each model are determined based on their precision and recall performance on the new_essay_val dataset. Below is a summary of the preliminary fine-tuning process.

Q8: What linguistic patterns distinguish AI-generated text from human text?
Ground Truth Answer: AI-generated text typically exhibits less variation in sentence length and tends to feature longer, more syntactically complex words compared to human writing . Additionally, AI models frequently use fewer stop words and adjectives, while distinctly favoring specific adverbs like "moreover," "particularly," and "additionally"
Expected Passage: Analyzing the sentence structure as depicted in Figure 4, the variation (standard deviation) of sentence length is compared between AIG and non-AIG texts, with AIG texts typically displaying less variation. This consistency across all datasets could be indicative of algorithmic constraints or stylistic limitations

inherent in AI-generated content. Zooming into the word level, Figure 5 demonstrates that AI-generated texts generally (both in the Kaggle and new_essay datasets) feature longer words. This observation suggests a potential tendency of AI models to select syntactically complex words.

Q9: How many competition essays were included in the new_essay dataset?
Ground Truth Answer: The new_essay dataset included 50 past winning essays sourced from five global competitions . These authentic essays were paired with GPT-3.5/4.0 generated content of similar lengths, creating a balanced dataset containing a total of 100 samples
Expected Passage: Under the competition essay category, an essay dataset included 50 past winning essays from five global competitions matched with GPT-3.5/4.0 generated content of similar lengths. This balanced dataset with a sum of 100 samples will be further random sampled in pairs for model finetuning and overall model evaluation. Furthermore, the test dataset comprised 25 pairs of essays from the John F. Kennedy “Profile in Courage” Essay Contest (2000-2021), alongside their GPT-4.0 generated counterparts, were utilized in deployment evaluation stage for ultimate deployment and product rollout.

Q10: What is the role of the test dataset in the evaluation pipeline?
Ground Truth Answer: The test dataset is utilized during the final deployment evaluation stage to validate the chosen model's effectiveness in real-world settings prior to its ultimate product rollout . By assessing the selected model with this data, developers can instill confidence in the system's ability to reliably deliver accurate results in practical applications
Expected Passage: The final model, selected based on superior performance, interpretability, and ease of implementation, undergoes a final deployment test with test_data. This test validates the model's effectiveness in real-world settings, instilling confidence in its ability to deliver reliable results in practical applications.

1.2 Apply Three Evaluation Metrics

#Q	Question (short)	Hit (0/1)	Faithfulness (0/1)	Relevance (0/1)
Q1	**Datasets used**	1	0	1
Q2	**Four features**	1	0	1
Q3	**What is DistilBERT**	1	1	1
Q4	**What is XLNet**	0	0	1

Q5	**Preprocessing steps**	0	1	1
Q6	**DistilBERT best precision/recall**	1	1	1
Q7	**Two-phase fine-tuning purpose**	0	0	1
Q8	**Linguistic patterns**	1	1	1
Q9	**Competition essay count**	1	1	1
Q10	**Role of test dataset**	1	0	1
Total		7/10	5/10	10/10

Metric 1 keyword choices:

1. "Kaggle", "HuggingFace", "competitions"
2. "length", "mean", "percentage"
3. "BERT", "faster", "efficient"
4. "understanding", "autoregressive", "pretraining"
5. "duplicate", "incorrectly", "extraction"
6. "precision", "recall", "fine-tuning"
7. "size", "specific", "structural"
8. "variation", "syntactically", "words"
9. "50", "authentic", "new_essay"
10. "evaluation", "validate", "effectiveness"

The evaluation scores indicate that the system's greatest strength is Answer Relevance (Metric 3), as the model consistently understands the prompt and directly addresses the user's questions. However, the evaluation scores indicate that the system's weakest area is Answer Faithfulness (Metric 2), as the LLM frequently hallucinates or introduces external claims not present in the retrieved context. While the initial retrieval step (Metric 1) performs relatively better at finding the correct passages, the generator fails to restrict its answers strictly to those provided texts. Furthermore, a clear failure pattern shows that when the retriever misses the relevant information, the model tends to guess the answer using its internal knowledge rather than admitting it does not know, which severely penalizes the faithfulness score.

Task 2: Chunking Strategy Experiment

2.2: Chunking Strategy Experiment Analysis

1. Retrieval Quality
The experiment results demonstrate a clear negative correlation between chunk size and retrieval hit rate. As the chunk size increased, the system's ability to retrieve the ground truth keywords decreased.

Chunk size	# Chunks produced	hit/10
256	192	8
512 (default)	90	7
1024	46	6

2. Answer Quality
The change in chunk size significantly impacted the LLM's answer quality, primarily by inducing hallucinations when retrieval failed.

- **Example 1 (Question 4 - "What is XLNet and how does it differ from traditional autoregressive models? "):** Under the small chunk size (256), the retriever successfully captured the exact mathematical definition, and the LLM correctly answered that XLNet uses a "permutation language modelling objective." However, when the chunk size increased to 512 and 1024, the retriever pulled broad summary tables instead of the specific definition. Consequently, the LLM hallucinated, falsely claiming in its answer that XLNet was "developed by OpenAI" and inventing incorrect architectural details.
- **Example 2 (Question 7 - "What is the purpose of the two-phase fine-tuning aproach?"):** At size 256, the system retrieved the exact sentence explaining the two phases and provided a perfectly faithful answer. At size 512, the retriever grabbed a chunk about computational limitations and dataset combinations instead. Without the correct context, the LLM fabricated a completely different explanation about "keeping hyperparameters fixed" which was not in the source document.

3. Trade-off Discussion

- Small chunks (256) create highly specific mathematical vectors. If a user asks a highly specific question, the dense vector of a small chunk containing the exact keyword will match perfectly, leading to high retrieval precision. However, if a concept requires a long, multi-paragraph explanation, a small chunk might forcefully cut the text in half, failing to provide the LLM with the complete context needed to formulate a comprehensive answer.

- Large chunks (1024) are excellent for keeping whole sections or long tables together without breaking the narrative flow. However, because a single large chunk contains many different topics and keywords, its vector representation becomes noisy. When a user asks a specific question, the large chunk's overall semantic meaning might not mathematically work well with the short query, causing the system to rank it lower and miss the relevant information entirely.

4. Recommendation

Based on the experimental results, I highly recommend **Chunk Size = 256** for this specific document. It achieved the highest retrieval hit rate (8/10), and because the user queries in this task are mostly factual and definition-based (e.g., "What is XLNet?", "What are the four features?"), the system benefits greatly from the high precision of small chunks. As proven by the answer quality examples, larger chunks introduced too much noise, leading to failed retrievals and forcing the LLM to generate severe hallucinations. Therefore, the small chunk strategy is the safest and most accurate choice for this application.